

Tree Distances II

Time limit: 1.00 s

Memory limit: 512 MB

You are given a tree consisting of n nodes.

Your task is to determine for each node the sum of the distances from the node to all other nodes.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print n integers: for each node $1, 2, \dots, n$, the sum of the distances.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 1 2 1 3 3 4 3 5	6 9 5 8 8